Safe Communication-Free MAPF

Chiawen Liao
chiawenl@andrew.cmu.edu

Thomas Chan
yuhsinch@andrew.cmu.edu

Wentao Cao
Email: wentaoc@andrew.cmu.edu

***Abstract*—Multi-agent path finding (MAPF) is a fundamental problem in multi-robot coordination. Centralized optimal solvers degrade in dense, large-scale settings, while decentralized learning-based approaches typically rely on inter-agent communication to resolve conflicts—an assumption that breaks down in subterranean environments, adversarial settings, or areas with heavy RF interference. We propose a communication-free variant of decentralized MAPF and address it through two complementary methods. First, we fine-tune MAPF-GPT on unsafe states mined using a local CS-PIBT safety shield, converting reactive shielding into proactive policy improvement. Second, we introduce an uncertainty-aware planner that leverages fixed priority orderings and a shared policy to simulate higher-priority agents, construct a cumulative occupancy risk map, and select actions that balance policy preference against collision risk. Preliminary benchmarking confirms that purely learning-based methods struggle in dense settings, but our PIBT-based data collection pipeline raised individual success rates from 0.35 to 1.00 in a 48-agent setup.**

I. Introduction & Motivation

Multi-agent path finding (MAPF) is a fundamental problem in multi-robot coordination: given a set of agents in a shared environment, the goal is to find collision-free paths from their start locations to their assigned goals. This problem arises in automated warehouses, autonomous vehicle coordination, and drone fleet management, where dozens to thousands of robots must navigate efficiently without colliding.

Finding an optimal MAPF solution is NP-hard. Optimal solvers such as CBS [11] perform well in sparse environments but degrade in dense, large-scale settings. Sub-optimal methods—including EECBS [7], LaCAM [8], PIBT [9], and Windowed-MAPF [15]—offer bounded-suboptimal or complete solutions within tractable time budgets, but remain centralized and slow as agent count scales.

Decentralized learning-based approaches such as MAPF-GPT [1] address scalability by having each robot independently plan from local observations. However, this introduces a coordination challenge: without knowledge of others' intentions, resolving conflicts is difficult. Works such as [12] and [13] mitigate this by re-centralizing coordination via inter-agent communication to run a conflict-resolution shield. This assumption is architecturally load-bearing: it breaks down in subterranean environments, adversarial settings with jamming, or dense warehouses with RF interference—precisely the safety-critical scenarios motivating MAPF.

In this work, we address a realistic variant of MAPF in which robots cannot communicate. We tackle this along two complementary directions: (1) learning a more conflict-aware policy via unsafe-state fine-tuning, and (2) designing a planning algorithm that reasons explicitly under uncertainty about unknown agent intentions.

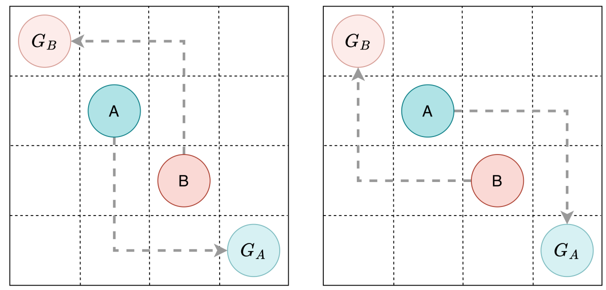

Fig. 1. A symmetric configuration where two independently planning agents have no basis to prefer one resolution over the other, making collisions likely.

II. Related Work

A. MAPF

1) Search-based methods: Search-based MAPF methods explicitly resolve inter-agent conflicts and are the dominant choice when formal guarantees are required. CBS [11] separates global conflict handling from low-level single-agent planning. LaCAM [8] improves scalability on large instances, while Windowed MAPF [15] shows that receding-horizon planning can retain completeness.

2) Learning-based methods: Learning-based methods replace expensive online search with reactive policies. PRIMAL2 [5] and SCRIMP [16] demonstrate that decentralized policies scale well at inference time. MAPF-GPT [1] and MAPF-GPT-DDG [2] show that large-scale imitation learning further improves generalization, though purely learning-based methods still lack formal safety or completeness guarantees under congestion.

3) Hybrid methods: Hybrid methods combine learned policies with search-based shielding. Augmenting a learned policy with CS-PIBT [13] effectively resolves local collisions, and even simple imitation learning becomes competitive once combined with such a planner [12]. These results highlight an open gap: existing hybrid methods rely on communication to aggregate agent plans before shielding, which fails in communication-denied settings.

B. Multi-agent interaction under uncertainty

Reasoning about other agents' behavior under uncertainty is a key challenge in decentralized settings. Sequential path planning (SPP) [3, 4] assigns strict priorities to agents, reducing joint planning complexity from exponential to linear, but assumes some state information is available. More recent approaches handle richer uncertainty: SN-CBFs [17] decompose multi-obstacle interactions into per-obstacle temporal sequences; contingency games [10] maintain a distribution over agent intents and plan contingent trajectories; and [6] uses adversarial RL to synthesize policies that account for the robot's evolving uncertainty at runtime. Our work draws on priority-based simulation in the spirit of SPP, but operates in a fully decentralized, communication-denied setting using only local observations and a shared policy.

III. Problem Formulation

We formalize communication-free decentralized MAPF as follows.

a) Environment.: An undirected graph $G = (V, E)$ represents the environment, where V is the set of traversable vertices and E connects adjacent vertices.

b) Agents.: N agents $\mathcal{A} = \{a_1, \ldots, a_N\}$ each have a distinct start $s_i \in V$ and goal $g_i \in V$.

c) Actions and observations.: Time is discretized. At each timestep t, agent a_i selects from $\{\textsc{Wait}, \textsc{Up}, \textsc{Down}, \textsc{Left}, \textsc{Right}\}$. Each agent observes obstacle structure, neighboring agent positions, and its own relative goal direction within a local radius r; we denote this o_i^t.

d) Assumptions.:

1) **Shared policy.** All agents execute the same policy $\pi : \mathcal{O} \to \mathcal{A}$ctions.
2) **Known priority ordering.** A fixed ordering $\sigma : \mathcal{A} \to \{1, \ldots, N\}$ is known to all agents prior to execution.
3) **Known map.** All agents have access to the full graph G.
4) **No communication.** Each agent acts using only o_i^t, π, σ, and G.

e) Constraints.: A valid execution must be vertex collision-free ($\text{pos}(a_i, t) \neq \text{pos}(a_j, t)$ for $i \neq j$), edge collision-free (no swap collisions), and complete (each agent reaches its goal in finite time).

f) Objective.: Minimize the sum-of-costs $\sum_{i=1}^{N} T_i$, where T_i is the first timestep agent a_i reaches g_i.

g) Distinction from prior work.: Unlike standard centralized MAPF or communication-dependent hybrid methods (e.g., MAPF-GPT + CS-PIBT), the safety and completeness constraints here must be satisfied *by the policy and algorithm alone*, without any inter-agent message passing at runtime.

IV. Proposed Approach

Our approach proceeds in two stages. In Stage 1, we improve the base MAPF-GPT policy by fine-tuning on unsafe states mined via a PIBT safety shield (Section IV-A). This produces a conflict-aware policy π^* that is more robust in dense settings. In Stage 2, we use π^* as the shared policy within an uncertainty-aware planning framework (Section IV-B). Because all agents execute the same policy, agent i can use π^* to simulate the future behavior of higher-priority neighbors and construct a local risk map. The fine-tuned policy thus serves a dual role: it directly improves each agent's action quality, and it provides a more accurate model for predicting other agents' behavior during risk-aware planning.

A. Learning a self-safe policy via unsafe state mining

One potential approach is to improve a pure learning-based MAPF policy by mining unsafe states and using them for supervised fine-tuning. We combine MAPF-GPT with a local CS-PIBT safety shield, and run the hybrid system on dense maps where failures are frequent. Whenever the shield is triggered, we record the corresponding state as an unsafe state and treat the shield's corrective action as expert supervision. We then fine-tune MAPF-GPT on these mined examples so that it can proactively avoid imminent collisions and local deadlocks from local observations. In this way, reactive shielding is converted into proactive policy improvement, reducing the dependence on external correction and improving robustness in dense scenarios.

B. Uncertainty-Aware Planning via Risk Map

Although the learned policy π^* encourages conflict-free behaviour, it cannot guarantee the absence of collisions. We leverage the fixed priority ordering and local observability within radius r (Section III) to simulate higher-priority agents and build a cumulative occupancy risk map for action selection.

1) Risk Map Construction: For agent i, let $P = \{p_1, \ldots, p_k\}$ be the higher-priority agents within radius r, ordered by priority. Since all agents share π^*, agent i simulates each agent in P for s timesteps. Actions with probability below ϵ are pruned and the distribution renormalised:

$$\tilde{\pi}(a \mid o) = \begin{cases} \pi^*(a \mid o) \big/ \sum_{a' : \pi^*(a' \mid o) \geq \epsilon} \pi^*(a' \mid o) & \text{if } \pi^*(a \mid o) \geq \epsilon, \\ 0 & \text{otherwise.} \end{cases} \tag{1}$$

For each simulated agent p_ℓ, we propagate marginal occupancy probabilities forward from its known position, initialising $r_{i_0 j_0 0}^{(p_\ell)} = 1$:

$$r_{ijt}^{(p_\ell)} = \sum_{(i',j') \in \mathcal{N}(i,j)} r_{i'j',\,t-1}^{(p_\ell)} \cdot \tilde{\pi}\big(a_{(i',j') \to (i,j)} \mid o_{p_\ell}^{t-1}\big) \tag{2}$$

Simulation proceeds in priority order; each agent avoids cells with high cumulative risk from higher-priority agents. The cumulative risk map aggregates per-agent occupancies via element-wise maximum:

$$R_{ijt} = \max_{p_\ell \in P} r_{ijt}^{(p_\ell)} \tag{3}$$

2) Implicit Risk Propagation: The raw risk map does not account for agents *passing through* cells en route to their goals. Using a distance-to-goal map d_{ij} (computed via backward Dijkstra), we propagate risk only toward the goal:

$$R_{xyt} \leftarrow \max\big(R_{xyt},\ R_{ijt} \cdot \alpha^{\,d_{ij} - d_{xy}}\big) \quad \text{if } d_{xy} < d_{ij} \tag{4}$$

where $\alpha \in (0, 1)$ is a decay factor.

TABLE I
BASELINE MAPF-GPT 2M PERFORMANCE WITHOUT SHIELDING.

Map Type	Agents	ISR	CSR	Collisions/Ep.
random	32	1.000	1.000	33.8
random	48	1.000	1.000	112.8
random	64	0.894	0.700	363.2
mazes	32	0.950	0.500	67.0
mazes	48	0.706	0.100	347.6
mazes	64	0.508	0.000	701.3
warehouse	64	0.983	0.800	79.9
warehouse	96	0.945	0.200	303.9
warehouse	128	0.749	0.100	992.6

3) Visited-State Penalty: To prevent livelock—agents oscillating between cells blocked by higher-priority neighbours—we add a penalty inspired by Anytime PIBT [14]. Let $V_i(x, y) = |\{t \in H_i : \text{pos}(a_i, t) = (x, y)\}|$ count how often agent i visited cell (x, y) in the last W timesteps.

4) Risk-Aware Action Selection: For each candidate action a leading to cell (x, y), the cost combines policy preference, collision risk, and revisitation penalty:

$$c(a) = -\lambda_1 \log \pi^*(a \mid o_i) + \lambda_2 \cdot R_{xy,\tau+1} + \lambda_3 \cdot V_i(x, y) \quad (5)$$

The agent samples from the softmin distribution $a^* \sim w(a)/\sum_{a'} w(a')$ where $w(a) = e^{-c(a)}$, preserving stochasticity to reduce the likelihood that multiple agents commit to the same conflicting action.

V. RESULTS

A. Baseline: MAPF-GPT 2M

We first evaluate the pretrained MAPF-GPT 2M policy as a purely learning-based decentralized MAPF baseline, without communication, explicit risk estimation, or runtime shielding. We report individual success rate (ISR), collective success rate (CSR), and average collisions per episode. ISR measures the fraction of agents that reach their goals, while CSR measures the fraction of episodes in which all agents succeed.

As results shown in Table I The baseline performs well in sparse or moderately dense random maps, achieving perfect CSR at 32 and 48 agents. However, its performance degrades significantly as the environment becomes denser or more topologically constrained. For example, on maze maps, CSR drops from 0.500 at 32 agents to 0.100 at 48 agents and 0.000 at 64 agents. A similar trend appears in the warehouse setting, where CSR falls from 0.800 at 64 agents to 0.100 at 128 agents. At the same time, the number of collisions increases rapidly with density, reaching 701.3 collisions per episode on 64-agent mazes and 992.6 collisions per episode on 128-agent warehouse maps.

These results show that the pretrained policy can generate reasonable local navigation actions, but it lacks an explicit mechanism for resolving dense multi-agent conflicts. In particular, narrow maze corridors and high-density warehouse layouts create repeated vertex and edge conflicts that the neural policy alone cannot reliably avoid. This motivates our two complementary directions: first, fine-tuning the policy to internalize collision-avoidance behavior, and second, adding an uncertainty-aware planner that explicitly reasons about future occupancy risk.

B. Fine-tuned Policy (π^*)

We next evaluate the fine-tuned policy π^* obtained through in-situ DAgger fine-tuning. Instead of training only on shield-modified unsafe states, which can cause catastrophic forgetting, the final training pipeline collects all observation-action pairs from shielded dense episodes. Unmodified actions preserve the original navigation behavior, while shield-modified actions provide supervision for collision avoidance. The resulting dataset contains 941,568 samples from 180 dense episodes across random, maze, and warehouse maps.

Fine-tuning substantially improves collision avoidance while preserving, and in many cases improving, goal-reaching performance. Across the evaluated scenarios, π^* reduces collisions by roughly 50% on average. The improvement is most visible in dense and structured environments: on 48-agent mazes, collisions drop from 347.6 to 126.0 per episode, while ISR improves from 0.706 to 0.906 and CSR improves from 0.100 to 0.400. On 128-agent warehouse maps, collisions decrease from 992.6 to 398.6, while ISR improves from 0.749 to 0.959 and CSR improves from 0.100 to 0.400. These results suggest that π^* has learned part of the shield's collision-avoidance behavior without losing the original policy's navigation capability.

We also evaluate the fine-tuned policy together with the CS-PIBT shield to measure whether π^* reduces the need for runtime intervention. This setting compares the original MAPF-GPT 2M with shield against π^* with the same shield.

TABLE II
SHIELD INTERVENTION RATE BEFORE AND AFTER FINE-TUNING.

Map Type	Agents	Original + Shield	π^* + Shield	Change
random	32	32.1%	16.3%	-49%
random	48	43.0%	33.3%	-23%
random	64	66.7%	55.9%	-16%
mazes	32	22.5%	16.4%	-27%
mazes	48	47.8%	40.9%	-14%
mazes	64	60.6%	62.1%	+2%
warehouse	64	27.5%	21.2%	-23%
warehouse	96	43.3%	37.5%	-13%
warehouse	128	56.7%	52.5%	-7%

Like results shown in Table II and Figure 2, the fine-tuned policy reduces shield intervention in almost all settings, showing that it proactively avoids many conflicts that previously required external correction. The average shield trigger reduction is about 20% when excluding the nearly deadlocked 64-agent maze case. In warehouse scenarios, π^* with shielding maintains or improves reliability while requiring fewer shield activations: for example, at 128 agents, CSR improves from 0.900 to 1.000 while the shield rate decreases from 56.7% to 52.5%. The main exception is the 64-agent maze scenario, where narrow corridors create persistent topological deadlocks and the shield intervention rate slightly increases. This suggests that fine-tuning is effective for reducing local collision conflicts, but dense maze-like environments may still require more global coordination or explicit deadlock resolution.

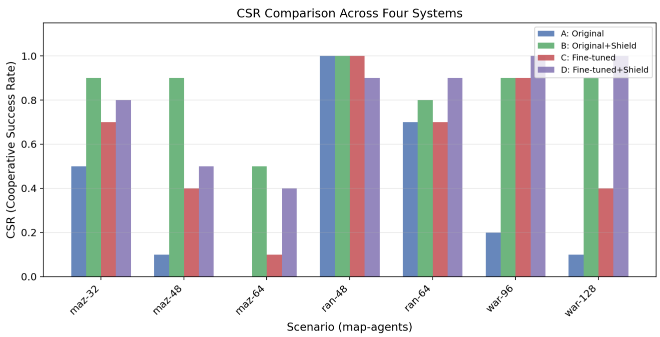

Fig. 2. CSR comparison across four systems: original MAPF-GPT 2M, original policy with CS-PIBT shield, fine-tuned policy π^*, and fine-tuned policy with shield. The shield improves reliability substantially, while fine-tuning provides standalone gains and further improves the shielded deployment setting in dense warehouse scenarios.

C. Uncertainty-Aware Planner

We evaluate three configurations on the POGEMA benchmark using the `01-random` and `02-mazes` evaluation suites with agent counts $N \in \{8, 16, 24, 32, 48, 64\}$: (1) **Vanilla**: the pre-trained MAPF-GPT-2M model with no wrapper; (2) **Vanilla + Safe**: the same model with the uncertainty-aware planner; (3) **Finetuned + Safe**: the fine-tuned policy π^* with the uncertainty-aware planner. Both planner configurations use $\lambda_3 = 0.3$, horizon $s = 3$, pruning threshold $\epsilon = 0.1$, decay $\alpha = 0.5$, $\lambda_1 = \lambda_2 = 1.0$, and visit window $W = 10$. Results are averaged across both suites.

Table III and Figure 3 show that the vanilla MAPF-GPT-2M policy degrades sharply with agent density.

1) Planner with vanilla policy: Applying the uncertainty-aware planner to the vanilla model yields large improvements, particularly in the mid-density regime. At 24 agents, CSR rises from 89% to 100%; at 32 agents, from 76% to 95%. At 64 agents, CSR increases from 32% to 56% and collisions decrease from 574 to 455 (21% reduction). The planner is most effective on random maps, where it achieves 86% CSR at 64 agents (vs. 58% vanilla), but also helps on mazes (26% vs. 6%). Runtime overhead is approximately 3× the vanilla baseline.

2) Planner with fine-tuned policy: Combining the fine-tuned policy π^* with the planner further improves results. CSR improves from 56% (vanilla+safe) to **68%** at 64 agents and from 84% to **85%** at 48 agents. Up to 24 agents, the system achieves **100% CSR** on both map types. Mean collisions per episode at 64 agents drop from 455 to **250** (45% reduction versus vanilla+safe, 56% versus vanilla alone).

The improvement is synergistic: the fine-tuned policy provides both better individual actions *and* a more accurate model for simulating higher-priority neighbors during risk-map construction. The gains are most pronounced on maze maps at high density: at 64 agents, CSR rises from 26% (vanilla+safe) to 45% (finetuned+safe). Runtime is comparable to vanilla+safe (22.3s vs. 22.4s at 64 agents), since the improved policy resolves conflicts earlier, offsetting the overhead of more accurate simulation.

TABLE III
PERFORMANCE COMPARISON AVERAGED OVER RANDOM AND MAZE SUITES. CSR AND ISR ARE REPORTED AS FRACTIONS; COLLISIONS ARE MEAN PER EPISODE.

N	Method	CSR	ISR	Coll.	Time (s)
32	Vanilla	0.76	0.96	56.4	2.1
	Van.+Safe	0.95	0.99	34.2	5.9
	FT+Safe	**0.93**	**0.99**	**18.8**	6.4
48	Vanilla	0.52	0.82	233.8	4.4
	Van.+Safe	0.84	0.97	138.4	12.7
	FT+Safe	**0.85**	**0.96**	**81.3**	13.3
64	Vanilla	0.32	0.71	573.7	6.8
	Van.+Safe	0.56	0.85	455.2	22.4
	FT+Safe	**0.68**	**0.90**	**249.5**	22.3

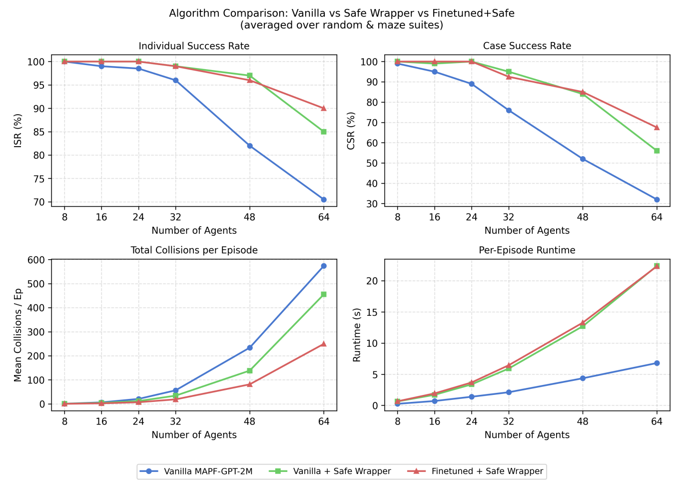

Fig. 3. ISR, CSR, collisions, and runtime across agent counts for the three configurations, averaged over random and maze suites.

VI. Conclusion

We presented a communication-free decentralized MAPF system targeting environments where inter-agent communication is denied. Our approach combines two complementary techniques. First, we collect hard states using a centralized CS-PIBT shield and fine-tune MAPF-GPT-2M on the shield's corrective actions, converting reactive shielding into proactive policy improvement. Second, we introduce an uncertainty-aware planner that exploits a shared policy assumption to simulate higher-priority neighbors, construct a cumulative risk map, and apply a visited-state penalty to avoid livelock.

Our results (Section V) show that fine-tuning alone substantially reduces collisions and improves completeness, while the planner alone provides strong gains in congested settings—raising CSR well above vanilla MAPF-GPT. The combination of both achieves the best overall performance, reaching 68% CSR at 64 agents compared to 32% for the vanilla baseline.

Several limitations remain. First, the system provides no formal safety or completeness guarantees: because simulation is limited to the local observation radius, deadlocks and collisions can still occur in highly congested scenarios. Second, the runtime of the uncertainty-aware planner grows with agent density, reaching approximately $3\times$ the baseline at 64 agents. Third, we assume perfect perception within the local field of view. In future work, we plan to evaluate the system under partial observability and noisy sensing, explore online priority adaptation to reduce deadlocks, and investigate whether larger fine-tuned models can further close the gap on maze maps.

References

[1] Anton Andreychuk, Konstantin Yakovlev, Aleksandr Panov, and Alexey Skrynnik. Mapf-gpt: Imitation learning for multi-agent pathfinding at scale, 2025. URL https://arxiv.org/abs/2409.00134.

[2] Anton Andreychuk, Konstantin Yakovlev, Aleksandr Panov, and Alexey Skrynnik. Advancing learnable multi-agent pathfinding solvers with active fine-tuning, 2025. URL https://arxiv.org/abs/2506.23793.

[3] Somil Bansal, Mo Chen, Jaime F Fisac, and Claire J Tomlin. Safe sequential path planning under disturbances and imperfect information. *arXiv preprint arXiv:1603.05208*, 2016.

[4] Mo Chen, Somil Bansal, Ken Tanabe, and Claire J Tomlin. Provably safe and robust drone routing via sequential path planning: A case study in san francisco and the bay area. *arXiv preprint arXiv:1705.04585*, 2017.

[5] Mehul Damani, Zhiyao Luo, Emerson Wenzel, and Guillaume Sartoretti. Primal$_2$: Pathfinding via reinforcement and imitation multi-agent learning - lifelong. *IEEE Robotics and Automation Letters*, 6(2):2666–2673, April 2021. ISSN 2377-3774. doi: 10.1109/lra.2021.3062803. URL http://dx.doi.org/10.1109/LRA.2021.3062803.

[6] Haimin Hu, Zixu Zhang, Kensuke Nakamura, Andrea Bajcsy, and Jaime F Fisac. Deception game: Closing the safety-learning loop in interactive robot autonomy. In *Conference on Robot Learning (CoRL)*, 2023.

[7] Jiaoyang Li, Wheeler Ruml, and Sven Koenig. Eecbs: A bounded-suboptimal search for multi-agent path finding, 2021. URL https://arxiv.org/abs/2010.01367.

[8] Keisuke Okumura. Lacam: Search-based algorithm for quick multi-agent pathfinding, 2022. URL https://arxiv.org/abs/2211.13432.

[9] Keisuke Okumura, Manao Machida, Xavier Défago, and Yasumasa Tamura. Priority inheritance with backtracking for iterative multi-agent path finding. *Artificial Intelligence*, 310:103752, September 2022. ISSN 0004-3702. doi: 10.1016/j.artint.2022.103752. URL http://dx.doi.org/10.1016/j.artint.2022.103752.

[10] Lasse Peters, Andrea Bajcsy, Chih-Yuan Chiu, David Fridovich-Keil, Forrest Laine, Laura Ferranti, and Javier Alonso-Mora. Contingency games for multi-agent interaction. *IEEE Robotics and Automation Letters*, 2024.

[11] Guni Sharon, Roni Stern, Ariel Felner, and Nathan R. Sturtevant. Conflict-based search for optimal multi-agent pathfinding. *Artificial Intelligence*, 219:40–66, 2015. ISSN 0004-3702. doi: https://doi.org/10.1016/j.artint.2014.11.006. URL https://www.sciencedirect.com/science/article/pii/S0004370214001386.

[12] Rishi Veerapaneni, Arthur Jakobsson, Kevin Ren, Samuel Kim, Jiaoyang Li, and Maxim Likhachev. Work smarter not harder: Simple imitation learning with cs-pibt outperforms large scale imitation learning for mapf, 2024. URL https://arxiv.org/abs/2409.14491.

[13] Rishi Veerapaneni, Qian Wang, Kevin Ren, Arthur Jakobsson, Jiaoyang Li, and Maxim Likhachev. Improving learnt local mapf policies with heuristic search, 2024. URL https://arxiv.org/abs/2403.20300.

[14] Rishi Veerapaneni, Jiaoyang Li, and Maxim Likhachev. Anytime single-step MAPF planning with anytime PIBT, 2025. URL https://arxiv.org/abs/2504.07841.

[15] Rishi Veerapaneni, Muhammad Suhail Saleem, Jiaoyang Li, and Maxim Likhachev. Windowed mapf with completeness guarantees. *Proceedings of the AAAI Conference on Artificial Intelligence*, 39(22):23323–23332, April 2025. ISSN 2159-5399. doi: 10.1609/aaai.v39i22.34499. URL http://dx.doi.org/10.1609/aaai.v39i22.34499.

[16] Yutong Wang, Bairan Xiang, Shinan Huang, and Guillaume Sartoretti. Scrimp: Scalable communication for reinforcement- and imitation-learning-based multi-agent pathfinding, 2023. URL https://arxiv.org/abs/2303.00605.

[17] Hongzhan Yu, Chiaki Hirayama, Chenning Yu, Sylvia Herbert, and Sicun Gao. Sequential neural barriers for scalable dynamic obstacle avoidance. 2023.